C-M-350C
August 31, 1977
SUPERSEDING
Int. Fed. Spec. C-M-00350B(Army-GL)
June 8, 1972 and
Fed. Spec. C-M-350A
January 10, 1962

FEDERAL SPECIFICATION

MILK, NONFAT, DRY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for nonfat dry milk.

1.2 Classification.

1.2.1 Types, styles and classes. Nonfat dry milk covered by this specification shall be of the following types, styles and classes, as specified (see 6.2):

Type I - Spray process (conventional):

- Style A - High heat
- Style B - Medium heat
- Style C - Low heat

Type II - Spray process (instantized):

- Style B - Medium heat
- Style C - Low heat
- Class 1 - Not fortified with vitamins
- Class 2 - Fortified with vitamins A, ascorbic acid, thiamine, B6

Type III - Roller process

- Style A - High heat

FSC 8910

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
UU-S-48	-	Sacks, Shipping, Paper
PPP-B-26	-	Bag, Plastic (General Purpose)
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-D-723	-	Drums, Fiber

Federal Standard:

FED-STD-123	-	Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-L-35078	-	Loads, Unit; Preparation of Nonperishable Subsistence

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Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-671	-	Sanitary Standards for Milk Evaporating and Drying Plants

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture

Directory of Meat and Poultry Inspection Program, Establishments and Officials
Brucellosis Eradication - Recommended Uniform Methods and Rules, Animal and Plant Health Inspection Service
U.S. Department of Agriculture, Publication 91-1
U.S. Standards for Condition of Food Containers
Uniform Methods and Rules - Bovine Tuberculosis Eradication

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products

U.S. Standards for Grades of Instant Nonfat Dry Milk
U.S. Standards for Grades of Nonfat Dry Milk (Spray Process)
U.S. Standards for Grades of Nonfat Dry Milk (Roller Process)
Dairy Plants Surveyed and Approved for USDA Grading Service

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U.S.D.A. Instructions:

918-103-1	Methods of Laboratory Analysis for Dry Whole Milk and Nonfat Dry Milk
918-103-2	Methods of Laboratory Analysis for Coliform Bacteria in Dry Milk Products
918-103-3	Methods of Laboratory Analysis for Isolation of Salmonella in Dry Milk
918-103-4	Methods of Laboratory Analysis for Whey Protein Nitrogen in Nonfat Dry Milk
918-103-7	Methods of Laboratory Analysis for Vitamin "A" in Dry Milk
918-103-8	Methods of Laboratory Analysis for Dispersibility of Nonfat Dry Milk

(Application for copies should be addressed to the Dairy Division, AMS, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

International Association of Milk, Food and Environmental Sanitarians, Inc.

3-A Sanitary Standards and Accepted Practices

(Application for copies should be obtained from the International Association of Milk, Food and Environmental Sanitarians, Inc., P.O. Box 701, Ames, IA 50010.)

Association of Vitamin Chemists

Methods of Vitamin Assay

(Application for copies should be addressed to the Association of Vitamin Chemists, 8946 Major Avenue, Morton Grove, IL 60653.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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3. *REQUIREMENTS*

3.1 Materials. Materials shall comply with the requirements of table I.

TABLE I. Material requirements

Item	Requirement
(1)	Milk. The raw milk used in the preparation of this product shall comply with all the quality requirements for No. 2 milk or better as specified in the General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products. The milk shall be drawn from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the US Department of Agriculture, or herds that have passed an annual tuberculosis test and meet USDA requirements for an individually certified herd, or from cows in herds located in: (a) A Modified Accredited Tuberculosis Area in accordance with USDA's Uniform Methods and Rules - Bovine Tuberculosis Eradication; and (b) Either (1) A certified Brucellosis-Free Area; or (2) Modified Certified Brucellosis Area; or (3) An area in the process of being certified by the USDA in accordance with USDA Animal and Plant Health Inspection Service Publication 91-1. The milk shall be practically free from colostrum, fresh, wholesome, and normal in appearance and odor and shall be subject to inspection by the procuring agency or its duly authorized representative.
(2)	Skim milk shall be obtained by the separation of whole milk conforming to the requirements of item 1, above. Prior to use, skim milk shall have a fresh, clean and pleasant flavor, and be normal in appearance.
(3)	Vitamins for type II, class 2 product shall comply with 3.3.

3.2 Processing. Processing shall be in accordance with table II.

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TABLE II. Processing

Item	Requirement
	Fluid milk or skim milk
(1)	All incoming milk or skim milk, unless processed within two hours, shall be cooled immediately to 45°F (7°C) or less.
(2)	Prior to condensing and drying, the skim milk to be used for processing shall be derived from plant separation of whole raw milk meeting the requirements of item 1, table I, and item 1 of this table. The skim milk shall be pasteurized by one of the following methods: (a) heating to at least 145°F (63°C) and holding at this temperature for not less than 30 minutes; (b) heating to at least 161°F (72°C) and holding at this temperature for not less than 15 seconds; or (c) any other pasteurization process recognized as equally efficient by the USPHS and approved by the appropriate State Authority. All pasteurization equipment, sanitary design, and construction standards shall conform to applicable 3-A Sanitary Standards and Accepted Practices.
(3)	Pasteurization of the raw whole or skim milk shall be accomplished as soon as practicable after entering the manufacturing plant and prior to the beginning of the condensing stage. Any interruption of the processing procedure for longer than one hour shall require recooling and holding of the milk at 45°F (7°C) or lower.
	Concentration of skim milk
(4)	Skim milk shall be pasteurized prior to evaporation in a manner which will yield a finished product of the desired level of whey protein nitrogen. Pasteurization may be accomplished by any method complying with 3-A Accepted Practices.
(5)	The product shall then be condensed in an evaporator (or vacuum pan) to a suitable solids level to permit proper drying. Vacuum levels and boiling points, and the use of single or multiple effect evaporators shall be in accordance with accepted good manufacturing practices and to permit compliance with the finished product requirements of 3.3. Any other form of concentration, as for example, ultrafiltration, shall be acceptable if it meets State and local health regulations.

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TABLE II. Processing (con.)

Item	Requirement
	Concentration of skim milk (cont'd)
(6)	The flow of concentrate from the evaporator to the dryer shall be directly through a closed pipeline or may be interrupted or retarded by: (a) passing through a pair of balance tanks to be used alternately every four hours with thorough cleaning and sanitizing between each use, and (b) using a storage tank for the concentrate, provided it passes through or over a continuous cooler, is cooled to 50°F (10°C) or lower upon entry to the tank, and is held at a temperature not to exceed 50°F (10°C) until entry into the final heat treatment unit prior to drying, and provided such concentrate is not held more than 30 hours. The final heat treatment shall incorporate a continuous flow of product which does not violate the provisions of item 3, above. In the event a by-pass line is included in the system, any product entering this line which is not immediately recirculated, shall be conducted immediately to a cooler and treated in the aforementioned manner.
	Drying
(7)	Spray dryers shall be of the continuous product removal type. Upon removal from the drying chamber, the product shall be cooled promptly to 110°F (43°C) or lower prior to packaging or bulk storage.
(8)	Type II product shall be produced by either a single spray drying process or by partially rehydrating and redrying a conventional spray dried nonfat dry milk in order to improve its dispersing and reliquefication characteristics. Nonfat dry milk, intended for partially rehydrating and redrying in the preparation of type II product shall be identified as to the name and location of the manufacturing plant and shall have been manufactured in a plant which meets the requirements of 3.6.
(9)	For type III product, the atmospheric or vacuum, single or double drum dryers, shall be used and operated in a manner which will yield a product complying with the requirements of 3.3.
(10)	Sifting prior to packaging shall be in accordance with recognized industry practices.

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TABLE II. Processing (con.)

Item	Requirement
	Drying (cont'd)
(11)	Vitamins, when specified in the contract, shall be incorporated so that the analytical levels specified in table III, item 2 are obtained in the finished spray-dried instantized product. The vitamins may be added in a premix to the condensed skim milk or the dried product prior to instantizing. No allowance has been made for loss in the quantities of vitamins specified, due to whichever procedure is adopted for their incorporation into the finished product.
(12)	Adulterants as defined in applicable US Grade standards shall not be added at any time before, during or after processing.

3.3 Finished product requirements. Finished product requirements shall, as a minimum, comply with US Extra Grade standards as set forth in the United States Standards for Grades of Nonfat Dry Milk (Spray Process, Instant, or Roller Process, as applicable). Supplemental requirements or exceptions, as applicable, shall be in accordance with table III.

Table III. Supplemental requirements or exceptions to US Extra Grade Standards

Item	Type I	Type II	Type III
(1) Whey protein nitrogen			
Style A, not more than	1.5 mg	-	1.5 mg
Style B, not less than	1.51 mg	1.51 mg	-
not more than	5.99 mg	5.99 mg	-
Style C, not less than	6.00 mg	6.00 mg	-
(2) Vitamins (applicable to type II, class 2 only)			
Vitamin A, not less than	-	4400 I.U. per 100 grams 1/	-
Ascorbic acid, not less than	-	70 mgs per 100 grams	

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TABLE III. Supplemental requirements or exceptions to US Extra Grade Standards (con.)

Item	Type I	Type II	Type III
Thiamine, not less than	-	1.8 mgs per 100 grams 1/	-
Vitamine B6, not less than	-	1.8 mgs per 100 grams 1/	-
(3) Microbiological			
Salmonella test 2/	Negative	Negative	Negative

1/ On a dry weight basis.

2/ Applicable to military agencies.

3.4 Age of product. The nonfat dry milk delivered under this specification shall be shipped within 60 days following initial drying of the fluid condensed skim milk. In case two-step processing procedures are employed for type II, this period shall begin immediately following the initial drying of the fluid condensed skim milk.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

ᵛ3.6 Plant qualification. The receiving stations, processing plant, premises, equipment, operations, personnel practices and sanitary practices used in the production and transportation of this item shall comply with the General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products and listed in the publication Dairy Plants Surveyed and Approved for USDA Grading Service. Optionally, for military agencies, the product shall originate, be processed, marked and handled in plants which meet the requirements of MIL-STD-671.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspections requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

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4.2 Plant qualification conditions. The product furnished under this document shall be unacceptable if not produced in plants complying with 3.6.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Condition inspection of containers. When inspection of containers for condition is performed by the US Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for the military, see 4.3.4).

4.3.2 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2.1 Inspection of milk and milk products.

4.3.2.1.1 Animal sources. Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in herds designated by the U.S. Department of Agriculture as tuberculosis-free and from herds designated brucellosis-free or in the process of being so designated. Determination of herd status regarding tuberculosis and brucellosis shall be made by contacting the nearest office of the Veterinary Services Office of Animal and Plant Health Inspection Service. (Location of this office may be ascertained by consulting the publication titled "Directory of Meat and Poultry Inspection Program, Establishments and Officials"). Nonconformance to either of the above requirements shall be cause for rejection of the lot.

4.3.2.1.2 Establishment sources. Examination shall be made to determine that all processed milk products have been manufactured, packaged and shipped by establishments approved as Military sources of supply as listed in the Directory of Sanitarily Approved Food Establishments for Armed Forces Procurement, or from plants listed by the U.S. Department of Agriculture in the document titled "Dairy Plants Surveyed and Approved for USDA Grading Service" and approved for the component products described in this specification. Nonconformance to the above requirements shall be cause for rejection of the lot or the involved quantity of finished product made therefrom, as applicable.

4.3.2.1.3 Examination of raw whole milk. Examination and testing of raw whole milk shall be made to determine compliance with the referenced requirements of table I, item 1, as concerns quality characteristics. Records of all required examinations and tests shall be maintained and made available for review. Failure to comply with one or more of the above referenced requirement(s), reflected by actual examination and tests or by examination of records shall be cause for rejection of the involved lot of the finished product.

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4.3.2.1.4 Examination of skim milk. Examination of skim milk shall be made to determine compliance with the requirements of table II, item 2 as concerns source and condition. This examination shall be performed by a review of plant records, laboratory reports or other valid documents. Failure to comply with the above referenced requirements, reflected by actual examination or by examination of records, shall be cause for rejection of the involved lot or finished product made therefrom, as applicable.

4.3.2.2 Testing of components. Testing of components described in this specification shall be as shown in table IV. Results shall be applicable to the lot average. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

TABLE IV. Testing of components (see 4.3.2)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to nearest	Test ref and rqmt para
Cans and lids 1/	One can w/lid	Cans and lids	S-1	Tin plate	0.01 lb/ base box	4.8 5.1.1
Foil laminated overwrap	1/2 sq. yd. (0.42 m^2) or one overwrap	Yards (meters) or overwraps	S-1	Thickness of aluminum foil lamina 2/ 3/	0.00001 inch (0.0001 mm)	5.1.2.3
				Basis weight	Pound/ream (g/m^2)	
Laminated foil bag	One bag	Bags	S-1	Thickness of each lamina 3/		
				Polyethylene	0.001 inch (0.01 mm)	5.1.2.4
				Aluminum foil	0.00001 inch (0.0001 mm)	
				Weight of polyethylene 3/	0.1 pound (0.01 kg)	

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TABLE IV. Testing of components (see 4.3.2) (con.)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to nearest	Test ref and rqmt para
				Basis weight of sulphite or sulphate paper	Pound/ream (g/m^2)	
Fiber drum bag	One drum bag	Drum bags	S-1	Thickness polyethylene bag 3/	0.001 inch (0.01 mm)	5.2.2.5
Bag to shipping sack	One bag	Bags	S-1	Thickness of polyethylene 3/	0.001 inch (0.01 mm)	5.2.2.6 5.2.3.3

1/ Exterior can coating, when required, and can label shall be tested in accordance with the applicable subsidiary specifications except that tests shall be performed on the same cans submitted for tin plate testing.

2/ See 5.1.2.3 for allowable tolerance.

3/ In lieu of testing the thickness of each laminae and weight of polyethylene (as applicable), determination of compliance to specified requirements may be ascertained by examination of labels, invoices or other valid documents. Nonconformance to thickness or weight requirements shall be cause for rejection of the lot.

4.3.3 Process examination. Unless otherwise specified (see 6.2), examination shall be performed during processing to ascertain compliance with the requirements of table II as concerns time, time intervals, temperature, pasteurization, drying equipment, vitamin fortification (when applicable), procedures and controls. Records of time and temperature shall be maintained. Nonconformance to one or more of the applicable requirements, as reflected by actual examination or by examination of time and temperature records, shall be cause for rejection of the involved lot of finished product.

4.3.3.1 In-process examination of drum and sack polyethylene liners. Examination of liners shall be made during packaging and packing to determine compliance with condition and dimensional requirements (as applicable) of 5.2.2.5 and 5.2.2.6. Fifteen liners of each lot, just prior to filling, shall be

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examined. The presence of objectionable odor, tears, holes or open seams or unclean condition in one or more liners shall indicate unacceptable liners and that portion of the finished product filled in the nonconforming lot of liners shall be rejected.

4.3.3.2 Examination of polyethylene and laminated bags and cartons (level B). Examination shall be performed to determine compliance with the applicable requirements of 5.1.2.2 through 5.1.2.4, as concerns type, style, class, variety, inner bridge (and dimensions), pour spout, overpack and overwrap material. This determination shall be ascertained by actual examination or by examination of pertinent labels, invoices or other valid documents. Failure to comply with one or more of the above referenced requirements shall be cause for rejection of the involved lot of containers or the finished product packaged therein.

4.3.4 Examination of end item. Examination of end item shall be performed in accordance with tables VI through VIII and 4.3.4.1 through 4.3.6. The examination criteria for the above referenced tables shall be as shown in table V. The acceptable quality levels (AQLs) shall be expressed as percent defective for tables VI and VII and defects per hundred units for table VIII.

TABLE V. End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLs	
				Major	Minor
VI	S-3	One primary container	Primary containers	--	2.5
VII	S-3	One primary container	Primary containers	0.65	--
VIII	S-3	One primary container	Primary containers	2.5	4.0

TABLE VI. Examination for net weight 1/

Category	Defect
Minor	
201	Net weight more than 2 percent under specified net weight. 2/
202	Net weight more than 1 percent under specified net weight. 3/

1/ Lot shall be rejected if sample data indicate lot average net weight is less than specified net weight.

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2/ Applicable to containers with more than 1 pound (0.45 kg) but less than 10 pounds (4.54 kg) net weight. Report results to the nearest 1/4 ounce (10 g).

3/ Applicable to containers with more than 10 pounds (4.54 kg) net weight. Report results to the nearest 4 ounces (100 g) in the range of over 10 pounds (4.54 kg) to 25 pounds (11.34 kg); and to the nearest 8 ounces (100 g) in the range of over 25 pounds (11.34 kg) to 50 pounds (22.68 kg).

TABLE VII. Examination for leakage 1/

Category	Defect
Major	
101	Steady progression of bubbles.

1/ Performed in accordance with 4.5.

TABLE VIII. Examination of bag-in-carton, overwrapped cartons or laminated bag (4, 4-1/2, or 5 lb. (1.81, 2.04 or 2.27 kg) fills - level B)

Category		Defect
Major	Minor	
101		Closure, as applicable, not heat sealed or closure incomplete.
102		Container, as applicable, not overpacked or overwrapped as specified. 1/
103		Evidence of delamination (see 5.1.2.4).
104		Sifter, tear or hole (inner and outer containers, as applicable).
105		Label nomenclature missing, incorrect or illegible.
106		Unclean exterior.
	201	Damaged or malfunctioning pour spout (see 5.1.2.3), when applicable.

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TABLE VIII. Examination of bag-in-carton, overwrapped cartons or laminated bag (4, 4-1/2, or 5 lb. (1.81, 2.04 or 2.27 kg) fills - level B) (con

Category		Defect
Major	Minor	
	202	Excessive air in bag.
	203	Excessive bulging in carton (see 5.1.2.3).
	204	Information other than nomenclature missing, incorrect or illegible.
	205	Not a single or double folded bottom. 1/

1/ Not applicable to level C container.

4.3.4.1 Examination of the finished product. Examination of the finished product shall be in accordance with U.S. Standards for Grades of Nonfat Dry Milk (Spray Process); U.S. Standards for Instant Nonfat Dry Milk; or U.S. Standards for Grades of Nonfat Dry Milk (Roller Process), as applicable. Any nonconformance with the finished product requirements in paragraph 3.3 shall be cause for rejection of the lot.

4.3.4.2 External examination of cans. When applicable, external examination of cans shall be performed in accordance with the applicable examination criteria of PPP-C-29.

4.3.4.3 Examination of can coating. When cans are required to be coated in accordance with TT-C-495, examination of coating shall be performed in accordance with that specification.

4.3.4.4 Examination for can labeling (Military). Examination for labeling shall be performed in accordance with the examination criteria of MIL-L-1497.

4.3.4.5 Examination of shipping containers. Examination of the filled and closed shipping container (except fiber drums and shipping sacks) shall be in accordance with the examination criteria in the appendix to PPP-B-636. In addition, the following defects shall be included in the table of examination: Major: Markings missing, incorrect or illegible; Minor: (1) Arrangement not as specified; (2) As applicable, missing or damaged case liner; (3) Case liner not protected as specified, when flaps of box are closed with metal fasteners;

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(4) When marked net weight of contents does not exceed 20 pounds (9.1 kg), box not as specified; (see 5.2.2.3). When applicable, reinforcement of the shipping container with other than pressure-sensitive adhesive, filament reinforced tape shall not be acceptable. When packing for level C is required to be in accordance with 5.2.3.2, examination shall be in accordance with the aforementioned specification, except that only the defects for marking and arrangement shall apply.

4.3.4.5.1 Examination of case liner. Case liners shall be examined in accordance with the examination criteria in the appendix to MIL-L-10547.

4.3.4.5.2 Examination of fiber drums. When applicable, examination of fiber drums shall be performed in accordance with the examination criteria of PPP-D-723. In addition, the following defects shall be included in the table of examination:

Major: (1) Markings missing, incorrect or illegible.
(2) Missing polyethylene bag or hole or tear in visible portion of top of bag.
(3) Unclean.

Minor: (1) Liner closure not as specified.
(2) Presence of excess air between product surface and top of bag.
(3) Marked net weight not as specified.

4.3.4.5.3 Examination of shipping sacks. When multiwall paper shipping sacks are required to be in accordance with UU-S-48, examination of the filled and closed sack shall be in accordance with the applicable quality assurance provisions contained in the appendix to that specification. In addition to the defects pertinent to this examination, the following defects shall be included: Major: (1) Required contractual markings missing, incorrect or illegible: (2) Sack damaged (e.g., tear, hole, open seam or broken stitches) adversely affecting serviceability; (3) Missing polyethylene bag; (4) Unclean liner in visible portion of top of sack; Minor: (1) Liner closure not as specified or tear,, hole or not sift proof; (2) Presence of excess air between product surface and top of sack; (3) Marked net weight not as specified.

4.3.5 Examination for age of product. Examination shall be made at time of shipment to determine compliance with 3.4. This determination shall be ascertained by examination of records, codes, marked date of pack or other valid means. Nonconformance to the above referenced requirement or the absence of any pertinent age records or manufacturing date codes shall be cause for rejection of the lot.

4.3.6 Examination of unit loads. Unitized loads shall be examined in accordance with the criteria of MIL-L-35078 and the applicable specification sheet for the type and class of load specified.

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4.4 Sampling procedure and acceptance criteria for testing of finished product. The finished product shall be tested for the physical, analytical, and microbiological requirements of 3.3, as applicable. Procedures for such testing shall be in accordance with 4.6, and all test references indicated below. Analytical test requirements shall be on a lot average basis and microbiological test requirements shall be on a unit basis. The sample unit for microbiological testing shall be one filled and sealed 1-pound (0.45 kg) can from production. For containers larger than 1-pound (0.45 kg), the sample unit shall be a 1-pound (0.45 kg) sample derived from each container greater than 1-pound (0.45 kg) net weight, utilizing aseptic technique. The sample size shall be the number of sample units indicated by inspection level S-1. Lot size shall be expressed in terms of pounds (kg). When applicable, the sample used for chemical, physical and vitamin testing shall be approximately 1-pound (0.45 kg), composited from the number of sample units utilized for the microbiological determination. Test results shall be reported in accordance with Methods of Laboratory Analyses for Dry Whole Milk and Nonfat Dry Milk Solids, Dry Buttermilk and Dry Whey - 918-103-1, 2, 3, 4, 7 and 8; ascorbic acid - nearest mg/100 g; thiamine and Vitamin B_6 - nearest 0.1 mg/100 g. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

4.5 Can leakage test. Can seams of the filled and closed cans shall be examined for leakage by submerging the can in water contained in a vacuum desiccator, Mead Tester, or equivalent device and drawing a vacuum of 10 inches of Mercury (atmospheric pressure 29.9 inches or 67.2kPa absolute pressure in metric equivalents) for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by air entrapped in the double seam are not considered as signs of leakage.

4.6 Chemical analysis (supplementary - see table III). Analyses shall be made in accordance with the methods cited below:

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Test	Source	Method
(1) Whey protein nitrogen	1/	Determination of Undenatured Whey Protein Nitrogen in Nonfat Dry Milk
(2) Vitamin A	2/	2/
(3) Ascorbic acid	3/	Photometric (Loeffler and Ponting modification)
(4) Thiamine	3/	Thiochrome
(5) Vitamin B_6	3/	Microbiological (Saccharomyces carlsbergensis)

1/ Methods of Laboratory Analysis for Whey Protein Nitrogen in Nonfat Dry Milk, 918-103-4.

2/ Methods of Laboratory Analysis for Vitamin A in Dry Milk, 918-103-7.

3/ Methods of Vitamin Assay of the Association of Vitamin Chemists, Inc.

4.7 Bacterial estimate. In accordance with referenced procedures in the applicable U.S. Standards for Grades of Nonfat Dry Milk (Spray, Instant and Roller Process), the bacterial estimate shall be made for the purpose of meeting U.S. Extra Grade requirements. In addition, Salmonella test procedures shall be conducted as outlined in the Methods of Laboratory Analysis for Isolation of Salmonella in Dry Milk, 918-103-3.

4.8 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Type I or II product, in the amounts and can sizes specified in table X, shall be filled into an open-top style, round metal can with soldered side seam and compound-lined, double-seamed ends. The can shall be made throughout from not less than 0.25-pound (0.52 kg) per base box electrolytic tin plate (see 4.8). The can shall be coated overall outside with a coating conforming to type I of TT-C-495. The hermetically sealed can shall not leak when examined in accordance with 4.5.

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TABLE X. Product type, can sizes, net weight, and number and arrangement of cans in snipping containers

Type	Net weight of product pounds (kg)	Can size	No. per case	Arrangement of cans in shipping containers		
				Length	Width	Depth
I	5 (2.27)	603 by 800	6	3	2	1
I	5 (2.27)	603 by 812	6	3	2	1
II	1 (0.45)	401 by 508 by 509	24	4	3	2
II	1 (0.45)	401 by 411	24	4	3	2
II	1 (0.45)	404 by 404	24	4	3	2
II	1 (0.45)	502 by 504	24	4	3	2
II	1 (0.45)	502 by 608	24	4	3	2
II	3 (1.36)	603 by 700	6	3	2	1
II	3 (1.36)	610 by 900	6	3	2	1

5.1.2 Level B. The product shall be packaged in accordance with 5.1.2.1, 5.1.2.2, 5.1.2.3, or 5.1.2.4 as specified (see 6.2). The following minus tolerances from specified net weights will be allowed in any individual container provided the average net weight of the container inspected in accordance with table VI is not less than the net weight specified: 2 percent for containers of one pound (0.45 kg) but less than 10 pounds (4.54 kg); one percent for containers 10 pounds (4.54 kg) and over.

5.1.2.1 Cans. Type I or II product shall be packaged in accordance with 5.1.1 except that cans with or without commercial exterior coating will be acceptable.

5.1.2.2 Bag-in-carton. A net weight of 4-1/2 pounds (2.04 kg) of type I product shall be packaged in a bag conforming to type I or II of PPP-B-26 constructed from 0.003 inch (0.076 mm) thick polyethylene. Excess air shall be removed from the bag and the bag shall be completely closed by heat sealing. The filled and sealed bag shall be overpacked in a paperboard carton constructed and closed in accordance with variety 1, style I, type A, class a or b of PPP-B-566.

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5.1.2.3 Overwrapped cartons. A net weight of 4 pounds (1.81 kg) of type II product shall be packaged in a paperboard carton constructed and closed in accordance with variety 1, style I, type A, class a or b of PPP-B-566 except that not less than 0.026 inch (0.66 mm) thick bleached paperboard shall be used. Each carton shall be provided with an inner bridge (minimum 7 inches (178 mm) in length) to lessen the bulge. When specified (see 6.2), each carton shall be equipped with a plastic or metal pull-out hinged pour spout located in the upper portion of the side panel. Each carton shall be furnished with an overwrap constructed from a foil laminated material. The aluminum foil lamina shall be 0.00035 inch (0.0089 mm) thick (± 10%). The completed overwrap shall consist of one or more paper laminae having a total combined basis weight of not less than 55 pounds per ream (24 x 36 - 500) (89g/m^2). The overwrap shall be completely closed by heat sealing.

5.1.2.4 Laminated bag. As specified (see 6.2), 4 or 5 pounds (1.81 or 2.27 kg) of type II product shall be packaged in a gusseted open-mouth laminated bag with a single or double folded bottom. The material shall be constructed from the following lamination (from inside to outside): 0.002 inch (0.05 mm) thick polyethylene laminated to 0.00035 inch (0.0089 mm) thick aluminum foil laminated with 7.5 pounds (3.40 kg) of polyethylene to 35 pound per ream (24 x 36 - 500) (57 g/m^2) sulphite or sulphate paper. Excess air shall be removed from the bag and the bag shall be completely closed by heat sealing.

5.1.3 Level C. Type I or II product shall be packaged in commercial cans, bags or paperboard cartons normally used for the product in net weights and container size specified in 5.1.1 or 5.1.2 for the particular product type and container used. Type III product shall be packaged in accordance with normal commercial practice.

5.2 Packing. Packing shall be in accordance with level A, B, or C, as specified (see 6.2 and 6.3). Fiberboard shipping containers may be top-opening or end-opening.

5.2.1 Level A. The cans of the product, in number and arrangement specified in table X, shall be packed on end in a snug-fitting shipping container constructed and closed in accordance with style RSC, V2s of PPP-B-636.

5.2.2 Level B. The product shall be packed in accordance with 5.2.2.1, 5.2.2.2, 5.2.2.5, or 5.2.2.6 as specified (see 6.2).

5.2.2.1 Cans. The cans of the product, in number and arrangement specified in table X, shall be packed on end in a snug-fitting fiberboard box constructed and closed in accordance with style RSC, V3c, V3s or V4s of PPP-B-636.

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5.2.2.2 Bags and cartons. Six bags or cartons of type II product shall be packed in a snug-fitting fiberboard box complying with 5.2.2.1 except that box style shall be RSC-L. Each shipping container shall be provided with a waterproof case liner conforming to MIL-L-10547 for subsistence items. When the flaps of the box are closed with metal stitches, the case liner shall be protected from the metal stitches by the use of a paperboard pad that completely separates the stitches from the case liner. The case liner shall be protected from stitches in the manufacturer's joint by placing a strip of pressure-sensitive tape over the stitches.

5.2.2.3 When specified (see 6.2), and when the net weight of product does not exceed 20 pounds (9.07 kg), the shipping container shall be constructed and closed in accordance with style RSC, W5s of PPP-B-636.

5.2.2.4 When specified (see 6.2 and 6.4), the shipping container shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.2.5 Fiber drums. When specified (see 6.2), 25 or 50 pounds (11.34 or 22.68 kg) of type I product shall be bulk packed in a fiber drum constructed and closed in accordance with type II, grade A, of PPP-D-723. The drum shall be provided with an inner bag conforming to type I or II of PPP-B-26 constructed from 0.003 inch (0.076 mm) thick polyethylene. The length of the polyethylene bag shall be not less than the length of the outer container plus one-half of the width of the outer container. The width of the polyethylene bag shall be capable of encircling the inside of the outer container without stretching or forming excessive folds. Excess air shall be expelled and the polyethylene bag shall be closed by heat sealing. Alternatively, the polyethylene bag shall be twisted and folded in a goose neck fashion and secured with tape, twine, or metal clips.

5.2.2.6 Multiwall paper shipping sack. When specified (see 6.2), 25 or 50 pounds (11.34 or 22.68 kg) of type I or type III product shall be bulk packed in a multiwall paper shipping sack complying with type I, style B, or type II, style A, construction 9 or 9 x of UU-S-48. The shipping sack shall be lined with an inner polyethylene bag constructed and closed in accordance with 5.2.2.5.

5.2.3 Level C. The cans of the product, in number and arrangement specified in table X, shall be packed on end in a snug-fitting shipping container constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure of PPP-B-636.

5.2.3.1 As applicable, 12 cartons of type I product or 6 cartons or 6 bags of type II product shall be packed in a snug-fitting fiberboard box complying with 5.2.3 except that box style shall be RSC-L for the type II product.

5.2.3.2 When specified (see 6.5) the product, in the amounts and quantities specified in the contract or order, shall be packed in commercial fiberboard boxes, shipping sacks, or drums in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

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5.2.3.3 When specified for Civil agencies (see 6.2), 100 pounds (45.36 kg) of nonfat dry milk shall be packed in a pack conforming to 5.2.3.2 which is fitted with an 0.003 inch (0.0762 mm) polyethylene film liner.

5.3 Unit loads. When specified (see 6.2) the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078, for type and class of load specified.

5.4 Labeling and marking.

5.4.1 Cans. Cans shall be labeled in accordance with MIL-L-1497 and as specified in 5.4.1.1 or 5.4.1.2, as applicable.

5.4.1.1 Type I, style A product. The label shall contain the information specified in 5.4.1.3 and as follows:

MILK, NONFAT, DRY
FOR RECOMBINED MILK
FOR BAKING OR COOKING

_____lb. net

Type I, Style A

Name and address of supplier

5.4.1.2 Type I, styles B and C, and type II product. The label shall contain the following information and the information specified in 5.4.1.3:

MILK, NONFAT, DRY
FOR BEVERAGE USE OR COOKING

_____lb. net

Type I, Style B

or

Type II (as applicable)

Name and address of supplier

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5.4.1.3 Additional labeling. In addition, the following information shall appear on the lid or the side of the can:

Direction for Use

Beverage - One pound nonfat dry milk and 3-3/4 quarts cool water to make 1 gallon.

Cooking - Thirteen ounces (2-1/2 to 3 cups) nonfat dry milk, 3-3/4 quarts water and 5 ounces shortening (10 tablespoons) for 1 gallon whole milk. In some recipes it is not necessary to use the shortening however, if used, it should be added with the other fat ingredients. The nonfat dry milk can be blended with the other dry ingredients and the water with other liquids. Add nonfat dry milk to 1/3 the amount of water; stir with a wire whip until dissolved and then blend in remaining water. Beverage milk should be made and chilled, if refrigeration is available. Handle prepared milk same as fresh milk. Keep cold.

NOTE: Since volume measurements vary, dry milk should be weighed rather than measured.

5.4.2 Bags, cartons and carton overwrap. The bags, cartons and carton overwrap shall be indelibly labeled with the information specified in 5.4.1.1 and 5.4.1.2, as applicable and with any other information as may be required by the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

5.4.3 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129 or FED-STD-123, as applicable. When shipping sacks or fiber drums are used as the shipping container, the information specified in 5.4.1.1 or 5.4.1.2, as applicable shall also appear on the exterior surface of the shipping sacks or drums. The additional information specified in 5.4.1.3 shall be printed on a sheet of white bond paper and the sheet shall be inserted into the shipping sack or fiber drum before closing.

5.4.4 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Intended use of product and technical information. The products covered by this specification are classified as nonperishable and have long, usable shelf-life in a dry state. Products have a high nutritive value and require less shipping weight than an equivalent amount of fluid milk. Storage in a cool location will aid in preserving an acceptable flavor. Products should not be stored in open or in damaged containers which would result in moisture pick-up or contamination with dirt, insects or rodents. They are intended for use as follows:

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Type I

Style A - In cooking and baking.

Style B - For direct use as a nonfat milk beverage, in recombined milk, or in cooking.

Style C - For direct use as a nonfat milk beverage, in cottage cheese, other cheese products, or other dairy products such as recombined milk.

Type II

Style B - For use in nonfat beverage milk or for cooking whenever improved reconstitution is desired over the use of other types or classes or when a less intense cooked flavor is the objective.

Style C - Used primarily in preparing nonfat beverage milk of high flavor quality.

The agglomerated condition of type II product makes it more bulky than type I powder and has about one-half the weight on an equal volume basis. Class 2 (fortified) is intended for use in Military combat rations or when special diets are indicated.

Type III - Used in cooking or baking as an ingredient in the manufacture of other prepared foods.

6.2 Ordering data. Purchasers should select the preferred options offered herein and include the following in procurement documents:

(a) Title, number and date of this specification.
(b) Type and style of product required and class for type II (see 1.2).
(c) When in-process examination is not required by Civil agencies (see 4.3.3).
(d) Levels of packaging and packing required (see 5.1 and 5.2).
(e) Size of can required for type I and II product (see 5.1.1).
(f) When plastic or metal pour spout is required (see 5.1.2.3).
(g) Whether 4 or 5 pounds (1.81 or 2.27 kg) of type II product is to be packaged in a laminated bag (see 5.1.2.4).
(h) Type of level B packing required (see 5.2.2).
(i) When packing in accordance with 5.2.2.3 or 5.2.2.4 is required.
(j) When 25 to 50 pounds (11.34 or 22.68 kg) of type I product are to be packed in a fiber drum (see 5.2.2.5).
(k) When 25 to 50 pounds (11.34 or 22.68 kg) of type I or III products are to be packed in a multiwall paper shipping sack (see 5.2.2.6).

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(l) Type and class of unit load when unit load is specified (see 5.3).
(m) When a commercial sack with a liner is required (see 5.2.3.3).

6.3 Military agencies. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/ AFR 71-6/MCO 4030.14D/DSAR 4145.7.

6.4 Packing specified in 5.2.2.4 is intended for transfer at sea operations or specific overseas operations.

6.5 Packing in 5.2.3.2 is intended for direct shipment from the supply source to the first receiving activity for use within CONUS.

6.6 Award of contract for the product specified in this document shall be limited to plants known to maintain the required sanitation conditions of 3.6 (also see 4.2).

6.7 Destination inspection (Civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS

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